Project: Worked examples
Title: EN 1993-1-1 LTB and N+M interaction
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1. Scope and method

Lateral-torsional buckling (6.3.2), flexural buckling (6.3.1) and the interaction of axial compression with bending (6.3.3) for a uniform doubly-symmetric I-section member on fork supports, class 1 to 3: M_cr, $\bar{\lambda}_{LT}$, χ_LT and M_b,Rd; χ_y, χ_z and N_b,Rd; the Annex B (or Annex A) interaction factors and both interaction ratios, with the governing check.
Critical moment (NCCI SN003 form): $M_{cr} = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 \cdot I_w/I_z + (kL)^2 \cdot G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$; $C_1 = 1.88 - 1.40\psi + 0.52\psi^2 \le 2.70$ (end moments, k = 1) unless entered LTB slenderness: $\bar{\lambda}_{LT} = \sqrt{(W_y \cdot f_y/M_{cr})}$, $W_y = W_{pl,y}$ (class 1, 2) or $W_{el,y}$ (class 3) 6.3.2.2: $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - 0.2) + \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \bar{\lambda}_{LT}^2)}) \le 1$ 6.3.2.3: $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \cdot \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \beta \cdot \bar{\lambda}_{LT}^2)}) \le 1$, $\le 1/\bar{\lambda}_{LT}^2$; optional $\chi_{LT,mod} = \chi_{LT}/f \le 1$, $f = 1 - 0.5(1 - k_c)[1 - 2(\bar{\lambda}_{LT} - 0.8)^2] \le 1$ Buckling resistance in bending: $M_{b,Rd} = \chi_{LT} \cdot W_y \cdot f_y/\gamma_{M1}$ Flexural buckling: $N_{cr} = \pi^2 EI/L_{cr}^2$, $\bar{\lambda} = \sqrt{(A \cdot f_y/N_{cr})}$, $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$, $N_{b,Rd} = \chi \cdot A \cdot f_y/\gamma_{M1}$ Interaction (6.3.3(4)): $N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1}) + k_{yy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{yz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$; $N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1}) + k_{zy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{zz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$ Factors k_ij: Annex B from the slenderness, $N_{Ed}/(\chi \cdot N_{Rk}/\gamma_{M1})$ and C_m ($C_m = 0.6 + 0.4\psi \ge 0.4$ for a linear diagram, or entered); Annex A from the μ, w, n_pl, a_LT and C_ij expressions

Ref: EN 1993-1-1:2005+A1:2014 6.3.1.2, 6.3.1.3 — Buckling curves; slenderness for flexural buckling
Ref: EN 1993-1-1:2005+A1:2014 6.3.2.2 — Lateral-torsional buckling curves: general case
Ref: EN 1993-1-1:2005+A1:2014 6.3.2.3 — Lateral-torsional buckling curves for rolled sections and equivalent welded sections
Ref: EN 1993-1-1:2005+A1:2014 6.3.3 — Uniform members in bending and axial compression
Ref: EN 1993-1-1:2005+A1:2014 Annex A, Annex B — Interaction factors k_ij: methods 1 and 2
Ref: NCCI SN003 Mcr expression — Elastic critical moment of a doubly-symmetric member
Ref: EN 1993-1-1:2022 8.3 — Stability of members (label only, see the edition note)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
edition	EN 1993-1-1 edition (2005+A1:2014 until the 2022 edition is adopted; the 2022 choice changes only the label, see the sheet note)	2005+A1:2014		spec: BS EN 1993-1-1:2005+A1:2014 with the UK National Annex	any
ltb_method	χ_LT method	6.3.2.3 (rolled and welded sections)		assumption: 6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd.	any
annex	Interaction factors	Annex B (method 2)		assumption: Annex B (method 2) chosen; Annex A can be selected for comparison.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
torsion	Member behaviour for Annex B	susceptible to torsional deformation (Table B.2)		assumption: An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).	any
f_y	Yield strength (for the thickness)	345	MPa	spec: BS EN 10025-2 S355, 16 mm < t ≤ 40 mm (flange 19.6 mm)	235 … 460 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 cl. 3.2.6	190 … 220 GPa
G	Shear modulus	81	GPa	spec: BS EN 1993-1-1 cl. 3.2.6: G = E/(2(1 + ν)) ≈ 81 GPa	70 … 90 GPa
gamma_M1	Partial factor γ_M1 (member buckling)	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.25
section_class	Cross-section class (class 4 needs effective properties and is outside this sheet)	1		assumption: Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).	any
section_type	Section type (selects the buckling curve of Table 6.4 / 6.5)	rolled I		spec: BS EN 10365 UB 457×191×98, hot-rolled	any
h	Overall depth h	467.2	mm	spec: BS EN 10365 UB 457×191×98	50 … 2500 mm
b	Flange width b	192.8	mm	spec: BS EN 10365 UB 457×191×98	30 … 1000 mm
A	Area A	125.265	cm^2	reference: S1 section sheet SEC-01, UB 457×191×98 (A = 125.265 cm²)	5 … 3000 cm^2
I_y	Second moment of area, major axis I_y (S1: I_xx)	45728	cm^4	reference: S1 section sheet SEC-01 (I_xx = 45727.99 cm⁴)	50 … 5000000 cm^4
I_z	Second moment of area, minor axis I_z (S1: I_yy)	2347.02	cm^4	reference: S1 section sheet SEC-01 (I_yy = 2347.02 cm⁴)	5 … 500000 cm^4
I_t	Torsion constant I_t (S1: J)	121.287	cm^4	reference: S1 section sheet SEC-01 (J = 121.287 cm⁴)	0.1 … 100000 cm^4

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I_w	Warping constant I_w	1.17554	dm^6	reference: S1 section sheet SEC-01 (I_w = 1.17554 dm⁶)	0.00001 … 100000 dm^6
W_pl_y	Plastic modulus, major axis (S1: W_pl_x)	2232.46	cm^3	reference: S1 section sheet SEC-01 (W_pl,x = 2232.46 cm³)	1 … 100000 cm^3
W_el_y	Elastic modulus, major axis (S1: W_el_x)	1957.53	cm^3	reference: S1 section sheet SEC-01 (W_el,x = 1957.53 cm³)	1 … 100000 cm^3
W_pl_z	Plastic modulus, minor axis (S1: W_pl_y)	378.904	cm^3	reference: S1 section sheet SEC-01 (W_pl,y = 378.904 cm³)	0.5 … 50000 cm^3
W_el_z	Elastic modulus, minor axis (S1: W_el_y)	243.467	cm^3	reference: S1 section sheet SEC-01 (W_el,y = 243.467 cm³)	0.5 … 50000 cm^3
L	Length between lateral restraints (fork supports)	5	m	drawing: FR-BM-12 rev B	0.1 … 100 m
k	Effective length factor k (end rotation about the minor axis)	1		assumption: Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).	0.5 … 1
k_w	Warping factor k_w	1		assumption: Free to warp at the restraints.	0.5 … 1
z_g	Load height z_g above the shear centre, towards the compression flange (positive destabilises)	0	mm	assumption: End-moment loading: no transverse load, so the load height does not enter.	-1000 … 1000 mm
C_2	C_2 (0 for end moments; enter it for transverse loads)	0		assumption: C_2 is 0 for end-moment loading.	0 … 1.5
ltb_curve	Buckling curve for χ_LT	auto (Table 6.4 / 6.5)		assumption: Curve from Table 6.5 for a rolled I with h/b > 2.	any
use_f	Apply the f modification to χ_LT (6.3.2.3(2))	yes		assumption: The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.	any
alpha_a0	Imperfection factor α, curve a0 (recommended 0.13)	0.13		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
alpha_a	Imperfection factor α, curve a (recommended 0.21)	0.21		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
alpha_b	Imperfection factor α, curve b (recommended 0.34)	0.34		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
alpha_c	Imperfection factor α, curve c (recommended 0.49)	0.49		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
alpha_d	Imperfection factor α, curve d (recommended 0.76)	0.76		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
lambda_LT0	$\bar{\lambda}_{LT,0}$ for 6.3.2.3 (recommended 0.4)	0.4		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.2 … 0.5
beta_LT	β for 6.3.2.3 (recommended 0.75)	0.75		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.5 … 1
moment_mode	Moment diagram given as	end moments		assumption: The moments are the member end moments.	any
N_Ed	Design axial force, TENSION positive and compression negative (as S2 N_e1 and S3; 6.3.3 applies to compression)	-400	kN	reference: S2 frame sheet FR-01, member 2, ULS combination (compression negative)	-100000 … 0 kN
M_y_Ed	Greatest design moment about y-y (ψ mode; S2 M_abs_e1 or S5)	300	kN*m	default	0 … 100000 kN*m
psi_y	ψ of the M_y diagram (ratio of the end moments, +1 uniform, −1 reverse symmetric)	0		default	-1 … 1
M_y_A	End moment A about y-y, signed (end-moment mode)	340	kN*m	reference: S2 frame sheet FR-01, member 2, end A	-100000 … 100000 kN*m
M_y_B	End moment B about y-y, signed, same sign convention as A (end-moment mode)	170	kN*m	reference: S2 frame sheet FR-01, member 2, end B (same side: single curvature)	-100000 … 100000 kN*m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
M_z_Ed	Greatest design moment about z-z	6	kN*m	assumption: Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).	0 … 100000 kN*m
psi_z	ψ of the M_z diagram	1		assumption: Uniform minor-axis moment (conservative).	-1 … 1
L_cr_y	Buckling length about y-y	5	m	drawing: FR-BM-12 rev B	0.1 … 100 m
L_cr_z	Buckling length about z-z	5	m	drawing: FR-BM-12 rev B	0.1 … 100 m
curve_y	Buckling curve about y-y (Table 6.2)	a		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about y-y	any
curve_z	Buckling curve about z-z (Table 6.2)	b		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about z-z	any

3. Assumptions

A1. Uniform doubly-symmetric I-section member on fork supports (free to warp and to rotate about the minor axis at the ends, restrained against twist); no mono-symmetric or tapered members, no cantilevers, no continuous-restraint effects. (Scope of S7; k and k_w below 1 model partial end restraint of the minor-axis rotation and of warping.)
A2. Class 1, 2 or 3 as stated by the user; class 4 effective properties are outside this sheet. (6.2.2.5 is not implemented.)
A3. C_1 from the ψ formula applies to end-moment diagrams with k = 1; for transverse loads, k ≠ 1 or other diagrams the user enters C_1, C_2 (and k_c, C_m) from the source in use. (No table of C_1 or C_2 values is shipped (reference records only).)
A4. The imperfection factors, β and $\bar{\lambda}_{LT,0}$ default to the recommended values and are a user-verifiable factor set; the National Annex in use may differ. (Record kind ec3.ltb-factors.)
A5. The 2022 edition choice changes the label printed on the sheet only. The arithmetic implemented is that of EN 1993-1-1:2005+A1:2014; clause numbers in the 2022 edition differ and must be checked by the user. (The differences between the editions are not implemented.)
A6. The Annex A and Annex B expressions are implemented as published; their coefficients are to be checked against the owner's copy of the standard. (Interaction factors for diagrams other than linear are entered, not derived.)
A7. ltb_method (χ_LT method) is taken as 6.3.2.3 (rolled and welded sections). (6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd.)
A8. annex (Interaction factors) is taken as Annex B (method 2). (Annex B (method 2) chosen; Annex A can be selected for comparison.)
A9. torsion (Member behaviour for Annex B) is taken as susceptible to torsional deformation (Table B.2). (An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).)

A10. section_class (Cross-section class (class 4 needs effective properties and is outside this sheet)) is taken as 1. (Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).)
A11. k (Effective length factor k (end rotation about the minor axis)) is taken as 1. (Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).)
A12. k_w (Warping factor k_w) is taken as 1. (Free to warp at the restraints.)
A13. z_g (Load height z_g above the shear centre, towards the compression flange (positive destabilises)) is taken as 0 mm. (End-moment loading: no transverse load, so the load height does not enter.)
A14. C_2 (C_2 (0 for end moments; enter it for transverse loads)) is taken as 0. (C_2 is 0 for end-moment loading.)
A15. ltb_curve (Buckling curve for χ_LT) is taken as auto (Table 6.4 / 6.5). (Curve from Table 6.5 for a rolled I with h/b > 2.)
A16. use_f (Apply the f modification to χ_LT (6.3.2.3(2))) is taken as yes. (The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.)
A17. moment_mode (Moment diagram given as) is taken as end moments. (The moments are the member end moments.)
A18. M_z_Ed (Greatest design moment about z-z) is taken as 6 kN*m. (Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).)
A19. psi_z (ψ of the M_z diagram) is taken as 1. (Uniform minor-axis moment (conservative).)

4. Working

[35] M_y_Ed = max(|M_A|, |M_B|) — 6.3.2
= max(|340|, |170|) kN·m
= 340 kN·m
Greatest design moment about y-y

[36] psi_y = □ = M_small/M_large (same sign: single curvature) — Table B.3
= 170 / 340
= 0.5
Ratio of the end moments

[37] C_1 = 1.88 − 1.40□ + 0.52□² ≤ 2.70 — NCCI SN003 / ENV 1993-1-1 Annex F
= □ = 0.5
= 1.31
C_1 for end moments (derived)

[38] Mcr = $C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 I_w/I_z + (kL)^2 G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$ — NCCI SN003
= 1.31 × π² × 210000 × 23470000 / (5000)² × [√((1/1)² × 1175500000000/23470000 + (5000)² × 81
1212900 / (π² × 210000 × 23470000) + (0 × 0)²) − 0]
= 808.38 kN·m
Elastic critical moment for lateral-torsional buckling

[39] W_y = W_pl,y (class 1, 2) — 6.3.2.2(1)
= plastic
= 2232.5 cm³
Section modulus used for bending about y

[40] lambda_LT = √(W_y·f_y/M_cr) — 6.3.2.2(1)
= √(2232.5 cm³ × 345 MPa / 808.38 kN·m)
= 0.9761
Non-dimensional slenderness for lateral-torsional buckling

[47] alpha_LT = imperfection factor, curve c — 6.3.2.3(1), Table 6.5
= Table 6.5, rolled, h/b = 2.4232
= 0.49
Imperfection factor for lateral-torsional buckling

[50] Phi_LT = 0.5[1 + □_LT(□̄_LT − □̄_LT,0) + □·□̄_LT²] — 6.3.2.3(1)
= 0.5 × [1 + 0.49 × (0.9761 − 0.4) + 0.75 × 0.9761²]
= 0.99843
Φ_LT

[51] chi_LT_0 = 1/(Φ_LT + √(Φ_LT² − Φ·λ̄_LT²)) ≤ 1, ≤ 1/λ̄_LT² 6.3.2.3(1)
= 1/(0.99843 + √(0.99843² − 0.75 × 0.9761²))
= 0.6537
Lateral-torsional reduction factor
[53] k_c = 1/(1.33 − 0.33ψ) Table 6.6
= ψ = 0.5
= 0.85837
k_c for a linear moment diagram (derived)
[54] f = 1 − 0.5(1 − k_c)[1 − 2(λ̄_LT − 0.8)²] ≤ 1 6.3.2.3(2)
= 1 − 0.5 × (1 − 0.85837) × [1 − 2 × (0.9761 − 0.8)²]
= 0.93358
Modification factor f
[55] chi_LT_mod = χ_LT/f ≤ 1 6.3.2.3(2)
= 0.6537 / 0.93358
= 0.70022
Modified lateral-torsional reduction factor
[56] chi_LT = χ_LT,mod 6.3.2.3
= as above
= 0.70022
Lateral-torsional reduction factor used
[57] Mb_Rd = χ_LT·W_y·f_y/γ_M1 6.3.2.1(3)
= 0.70022 × 2232.5 cm³ × 345 MPa / 1
= 539.3 kN·m
Design buckling resistance moment
[58] u_LTB = M_y_Ed / Mb_Rd
= 340 kN·m / 539.3 kN·m
= 0.63044
Lateral-torsional buckling utilisation
[61] alpha_y = imperfection factor, curve a 6.3.1.2
= factor set
= 0.21
Imperfection factor about y
[62] alpha_z = imperfection factor, curve b 6.3.1.2
= factor set
= 0.34
Imperfection factor about z
[63] N_Rk = A·f_y 6.3.1.1(3)
= 125.27 cm² × 345 MPa
= 4321.6 kN
Characteristic axial resistance
[64] N_cr_y = π²E·I_y/L_cr,y² 6.3.1.2
= π² × 210000 × 457280000 / 5000²
= 37911 kN
Elastic critical force about y
[65] N_cr_z = π²E·I_z/L_cr,z² 6.3.1.2
= π² × 210000 × 23470000 / 5000²
= 1945.8 kN
Elastic critical force about z
[66] lambda_y = √(A·f_y/N_cr,y) 6.3.1.3
= √(4321.6 / 37911)
= 0.33763
Non-dimensional slenderness about y

[67] lambda_z = √(A·f_y/N_cr,z) 6.3.1.3
= √(4321.6 / 1945.8)
= 1.4903
Non-dimensional slenderness about z
[68] chi_y = 1/(Φ + √(Φ² − λ̄²)) ≤ 1 6.3.1.2
= Φ = 0.21, λ̄ = 0.33763
= 0.96853
Reduction factor about y
[69] chi_z = 1/(Φ + √(Φ² − λ̄²)) ≤ 1 6.3.1.2
= Φ = 0.34, λ̄ = 1.4903
= 0.34582
Reduction factor about z
[70] N_b_y_Rd = χ_y·A·f_y/γ_M1 6.3.1.1(3)
= 0.96853 × 4321.6 / 1
= 4185.7 kN
Buckling resistance about y
[71] N_b_z_Rd = χ_z·A·f_y/γ_M1 6.3.1.1(3)
= 0.34582 × 4321.6 / 1
= 1494.5 kN
Buckling resistance about z
[72] N_b_Rd = min(N_b_y_Rd, N_b_z_Rd)
= min(4185.7 kN, 1494.5 kN)
= 1494.5 kN
Flexural buckling resistance (smaller of the two axes)
[73] M_y_Rk = W_y·f_y 6.3.3
= 2232.5 cm³ × 345 MPa
= 770.2 kN·m
Characteristic bending resistance about y
[74] M_z_Rk = W_z·f_y (W_pl,z class 1, 2; W_el,z class 3) 6.3.3
= 378.9 cm³ × 345 MPa
= 130.72 kN·m
Characteristic bending resistance about z
[75] C_my = 0.6 + 0.4ψ_y ≥ 0.4 (or entered) Table B.3
= ψ_y = 0.5
= 0.8
Equivalent uniform moment factor, y
[76] C_mz = 0.6 + 0.4ψ_z ≥ 0.4 (or entered) Table B.3
= ψ_z = 1
= 1
Equivalent uniform moment factor, z
[77] C_mLT = as C_my (or entered) Table B.3
= C_my
= 0.8
Equivalent uniform moment factor for lateral-torsional buckling
[78] k_yy = Annex B: C_my(1 + (λ̄_y − 0.2)n_y) ≤ C_my(1 + 0.8n_y) (class 1, 2) Annex B Table B.2
= n_y = 0.095565
= 0.81052
Interaction factor k_yy
[79] k_yz = Annex B: 0.6k_zz (class 1, 2), k_zz (class 3) Annex B Table B.2
= as above
= 0.82482
Interaction factor k_yz

[80] k_zy = Annex B: Table B.1 0.6k_yy / 0.8k_yy; Table B.2 from χ_z, n_z and C_mLT Annex B Table B.1
= n_z = 0.26764
= 0.95134
Interaction factor k_zy
[81] k_zz = Annex B: C_mz(1 + (2χ_z − 0.6)n_z) ≤ C_mz(1 + 1.4n_z) (class 1, 2) Annex B Table B.2
= as above
= 1.3747
Interaction factor k_zz
[82] u_N_y = N_Ed/(χ_y·N_Rk/γ_M1) 6.3.3(4)
= 400 kN / (0.96853 × 4321.6 kN / 1)
= 0.095565
Axial term about y
[83] u_N_z = N_Ed/(χ_z·N_Rk/γ_M1) 6.3.3(4)
= 400 kN / (0.34582 × 4321.6 kN / 1)
= 0.26764
Axial term about z
[84] u_M_y = M_y,Ed/(χ_LT·M_y,Rk/γ_M1) 6.3.3(4)
= 340 kN·m / (0.70022 × 770.2 kN·m / 1)
= 0.63044
Major-axis bending term
[85] u_M_z = M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 6 kN·m / (130.72 kN·m / 1)
= 0.045899
Minor-axis bending term
[86] u_61 = N_Ed/(χ_y N_Rk/γ_M1) + k_yy·M_y,Ed/(χ_LT M_y,Rk/γ_M1) + k_yz·M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 0.095565 + 0.81052 × 0.63044 + 0.82482 × 0.045899
= 0.64441
Interaction, equation 6.61
[87] u_62 = N_Ed/(χ_z N_Rk/γ_M1) + k_zy·M_y,Ed/(χ_LT M_y,Rk/γ_M1) + k_zz·M_z,Ed/(M_z,Rk/γ_M1) 6.3.3(4)
= 0.26764 + 0.95134 × 0.63044 + 1.3747 × 0.045899
= 0.9305
Interaction, equation 6.62
[88] u_interaction = max(u_61, u_62) 6.3.3(4)
= max(0.64441, 0.9305)
= 0.9305
Greatest interaction ratio
[89] u_buckling = N_Ed/N_b,Rd 6.3.1.1
= 400 kN / 1494500 kN
= 0.26764
Flexural buckling utilisation
[90] u_member = max(max(u_interaction, u_LTB), u_buckling)
= max(max(0.9305, 0.63044), 0.26764)
= 0.9305
Governing utilisation of the member

5. Results

SYMBOL	VALUE	DESCRIPTION
edition_used	"EN 1993-1-1:2005+A1:2014"	EN 1993-1-1 edition
psi_y	0.5	ψ of the M_y diagram
M_y_Ed	340 kN·m	Greatest design moment about y-y
C_1	1.31	C_1

SYMBOL	VALUE	DESCRIPTION
Mcr	808.38 kN·m	M_cr
lambda_LT	0.9761	λ̄_LT
chi_LT	0.70022	χ_LT
Mb_Rd	539.3 kN·m	M_b,Rd
N_Rk	4321.6 kN	N_Rk
N_cr_y	37911 kN	N_cr,y
N_cr_z	1945.8 kN	N_cr,z
lambda_y	0.33763	λ̄_y
lambda_z	1.4903	λ̄_z
chi_y	0.96853	χ_y
chi_z	0.34582	χ_z
N_b_y_Rd	4185.7 kN	N_b,y,Rd
N_b_z_Rd	1494.5 kN	N_b,z,Rd
N_b_Rd	1494.5 kN	N_b,Rd
k_yy	0.81052	k_yy
k_yz	0.82482	k_yz
k_zy	0.95134	k_zy
k_zz	1.3747	k_zz
u_LTB	0.63044	LTB utilisation
u_buckling	0.26764	Buckling utilisation
u_61	0.64441	Equation 6.61
u_62	0.9305	Equation 6.62
u_interaction	0.9305	Interaction utilisation
u_member	0.9305	Member utilisation
alpha_LT	0.49	α_LT
Phi_LT	0.99843	Φ_LT
chi_LT_0	0.6537	χ_LT before the f modification
k_c	0.85837	k_c
f	0.93358	f
W_y	2232.5 cm³	W_y
M_y_Rk	770.2 kN·m	M_y,Rk
M_z_Rk	130.72 kN·m	M_z,Rk
C_my	0.8	C_my
C_mz	1	C_mz
C_mLT	0.8	C_mLT

SYMBOL	VALUE	DESCRIPTION
u_N_y	0.095565	Axial term, y
u_N_z	0.26764	Axial term, z
u_M_y	0.63044	Bending term, y
u_M_z	0.045899	Bending term, z
governing	"interaction 6.62 (0.931)"	Governing check

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Interaction, equation 6.62	u_62 = 0.9305	u_62 ≤ 1	0.931	6.3.3(4)
PASS	Interaction, equation 6.61	u_61 = 0.64441	u_61 ≤ 1	0.644	6.3.3(4)
PASS	Lateral-torsional buckling M_y,Ed ≤ M_b,Rd	u_LTB = 0.63044	u_LTB ≤ 1	0.63	6.3.2.1
PASS	Flexural buckling N_Ed ≤ N_b,Rd	u_buckling = 0.26764	u_buckling ≤ 1	0.268	6.3.1.1

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: E, G, f_y (Material properties); gamma_M1 (EN 1993-1-1 National Annex factors); alpha_a0, alpha_a, alpha_b, alpha_c, alpha_d, lambda_LT0, beta_LT (Imperfection factors and 6.3.2.3 parameters); C_2 (C_1, C_2 and k_c for the moment diagram). Enter each as one so its source, edition and licence travel with it.
WARNING — input.M_y_Ed: M_y_Ed (Greatest design moment about y-y (ψ mode; S2 M_abs_e1 or S5)) was not entered; the template default 300 kN*m is used.
WARNING — input.psi_y: psi_y (ψ of the M_y diagram (ratio of the end moments, +1 uniform, −1 reverse symmetric)) was not entered; the template default 0 is used.

8. Verification

Template struct.ltb-interaction@1.0.0 Content hash f1fbb73d8a55eef2 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0